

MATH3171 (Mathematical Biology III)
YEAR 2025–26, EPIPHANY TERM

PROBLEMS CLASS 5

1 Review: Simple disease dynamics (Again! Shouldn't we be done with this already...?)

Recall the SIS model from Michaelmas:

$$\begin{aligned}\frac{dS}{dt} &= \delta I - \beta SI, \\ \frac{dI}{dt} &= \beta SI - \delta I.\end{aligned}$$

- (a) If $S(t)$ is the number of susceptible individuals at time t , and $I(t)$ the number of infected, what do the parameters δ and β represent? What assumptions are inherent to this model which makes it different from other compartmental epidemiological models you've seen (e.g. SIR)?
- (b) Show that the quantity $N(t) = S(t) + I(t)$ does not change in time. How would you compute $N(t)$ if you knew the initial susceptible and infected populations?
- (c) Use the fact that N is a constant to write an equation for the rate of change of the proportion of infected individuals, $p = I/N$. That is, find an equation for dp/dt in terms of just p that looks like

$$\frac{dp}{dt} = (B - \delta)p - Bp^2,$$

where you should specify B in terms of model parameters.

- (d) Compute the equilibria of this equation for p , and determine their permissibility and linear stability.

2 Differences that models make

Now let's consider a discrete-time variant of the SIS model where an infection starts on the first day with S_0 susceptible individuals and I_0 infected individuals. We model the number of people who are susceptible and infected on day n by:

$$\begin{aligned}S_n &= S_{n-1} + \delta I_{n-1} - \beta S_{n-1} I_{n-1}, \\ I_n &= I_{n-1} + \beta S_{n-1} I_{n-1} - \delta I_{n-1},\end{aligned}$$

where the (always positive) parameters now represent rates of infection, recovery etc within one day (we have already nondimensionalized one day to be the difference between n and $n - 1$).

- (a) What terms are different in this model from its continuous-time (ODE) counterpart? Explain in words what each term in each equation represents.

- (b) Define a discrete-time version of the total population and show that it will not change in time (that is, N is a number which doesn't depend on the day n).
- (c) Using the above constant total population, reduce this model to a single difference equation for the total proportion of infected individuals, $p_n = I_n/N$ of the form,

$$p_n = R_0 p_{n-1} - B p_{n-1}^2,$$

where you should define both R_0 and B in terms of the original parameters in the model.

- (d) Compute the equilibria p^* of this model. Determine the permissibility and linear stability of any of these equilibria as the parameters R_0 and B vary.

3 Discrete SIR

Now let's consider a discrete-time variant of the SIR model of the form:

$$\begin{aligned} S_n &= S_{n-1} - \beta S_{n-1} I_{n-1}, \\ I_n &= I_{n-1} + \beta S_{n-1} I_{n-1} - \delta I_{n-1}, \\ R_n &= R_{n-1} + \delta I_{n-1}, \end{aligned}$$

where $\beta > 0$ and $\delta > 0$.

- (a) Define a total population N and show that it must be constant.
- (b) What are the equilibria of this model? How do they relate to the total population N ?
- (c) Compute the net rate of new infected individuals each day. Use this to determine a condition as a function of the parameters and S_{n-1} such that the net rate of infectives on day n is decreasing.
- (d) Assuming we have an initial population composed of $N = S_0 + I_0$ infected and susceptible people, and that we can control the rate at which infected people recover δ . What is the smallest value that we could set δ to be equal to in order to prevent the total number of infected people from ever rising? Interpret this result for a fixed number of initially infected people I_0 for different population sizes N .